FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

SJ

Singha Junchan

สิงหา จุลจันทร์ / Pathum Thani, Thailand

I build access control for cloud and AI systems.

- **1st in Thailand**: national champion, Thailand Olympiad in Informatics (TOI18), 2022.
- **Best Paper Award**, IEEE JCSSE 2026, for **U-HAP**, Kubernetes authorization (second author).
- **9 security research projects** on **Kubernetes**, **zero trust**, **zero-knowledge proofs**, **post-quantum** cryptography and **AI agents**; first author of **FEDAC** (under review, IEEE TDSC).
- **Security intern, Allianz Ayudhya**: learned **security governance and risk** from the CISO and built **two AI agents for third-party risk**.
- **Team lead, TU Alert**: campus incident reporting in **LINE**, built for Thammasat University (launching soon).
- Final-year **Computer Engineering**, SIIT, Thammasat University: **GPA 3.67**, **full scholarship**, graduating **May 2027**.

Scan or click for the full portfolio, code links and contact.

singhaju.github.io

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Open to: security internships from January 2027, in Thailand or abroad full-time roles from May 2027

COMPETITIVE PROGRAMMING
#1 in Thailand
TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for **Thammasat University**; **I lead the team**; security built in; launching soon

RESEARCH
9 research projects
2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY
2 internships
Allianz Ayudhya (security governance and risk, with the CISO) and **Amity Solutions** (LLM chatbots)

LEADERSHIP AND TEACHING
11 student events
led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION
ISC2 Certified
Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 What I bring

Security engineering, end to end
I **design security mechanisms and build them**: a zero-trust framework with a zero-knowledge verifier (**first author**) and a Kubernetes authorization webhook (**Best Paper**, second author).

Identity, access and governance
Who may do what, checked on every request and **provable later**. Growing into **governance**: policy, third-party risk and audit.

Cloud-native by default
I test ideas on **real cloud deployments**: Kubernetes on **AWS EKS**, prototypes on **EC2**, containers with **Docker**.

AI agents, and how to limit them
I **build AI agents** that do useful work, and **research controls** that keep them inside what a human allowed, even under attack.

Leads, teaches and explains
I **start things and run teams**: TU Alert, 11 student events, two coaching platforms. I explain hard ideas simply.

02 Education and leadership

Education

B.Eng. in Computer Engineering — Aug 2023 to May 2027 (expected)
Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University GPA 3.67 / 4.00, first-class honours expected.

Leadership

11 student events organised · ~500 participants at SIIT First Meet 2024 · 100 staff at SIIT First Meet 2024

President, Sub Student Committee 32, SIIT — 2024 to present
Leads a committee that represents the student body and organises academic and social events.

Lead organiser, SIIT First Meet 2024 — 2024
About 500 participants and 100 staff. Organiser of 10 more student events, 2023 to 2025.

Project manager and product owner, Novel Nest — Team project
Led a team of four to build a digital library and creative hub for readers and writers.

03 Experience

Team Lead and Developer, TU Alert

● Launching soon

Thammasat University (client project) / Rangsit campus, Pathum Thani — 2026

- **What it is:** a campus incident-reporting system that runs **inside LINE**, built for **Thammasat University** (launching soon). A student reports a problem with photos and a map pin, and a rule sends it to the office that owns it. **I lead the team** and write code.
- **LINE integration:** a student **LINE Official Account** (rich menu, Flex message cards, chat commands, star ratings by Quick Reply), a **group bot** that alerts each office's LINE group and answers /queue and /summary, and two **LIFF apps** with LINE sign-in checked on the server.
- **Student side:** report with photo, video and a campus map pin (kept on the phone until sent), track the status, chat with the office, rate the result, reopen it if not fixed. **Staff side:** a phone app and a web portal to accept, transfer, resolve and assign; SLA and analytics for heads; emergency broadcast to students.
- **Security built in:** four roles checked on every request, second-step staff sign-in (authenticator app, SMS or LINE approval), kill switches, an audit log, PDPA data export and erasure; **Semgrep, gitleaks and OSV** scans in CI.

Secure web development · Role-based access control · Data retention and erasure · Team leadership · TypeScript · Next.js · Hono · Bun · Prisma · MariaDB · Redis · LINE LIFF · Semgrep · gitleaks

Information Security Intern

Allianz Ayudhya / Bangkok, Thailand — Jun to Jul 2026

- **What it is:** the Information Security team's **third-party risk management (TPRM)**: checking a supplier's security when the company buys, and answering clients' **due-diligence questionnaires** when it supplies. **Mentored by the CISO**, I learned how **security governance and risk management** work in practice.
- **Designing AI into the work:** mapped the manual questionnaire process end to end with the team to find where AI helps most, and set what every AI answer must carry: **evidence, a reason and a confidence score**, so a person can check it fast.
- **Questionnaire agent:** reads policy and evidence documents, drafts each answer, and runs **gap and fit-gap analysis** to show where current controls fall short. **Guidance agent:** teaches business owners each step, tracks their progress, checks their inputs, **suggests what to do next and flags** items to review.
- Tested on **real, anonymized cases** with the team; demonstrated on 6 Jul and **presented both** on 31 Jul 2026.

Information security governance · Risk management · Third-party risk management · Due-diligence questionnaires · Gap and fit-gap analysis · AI agents

Founder, Curriculum Author and Lead Instructor

TOI Tutor and ABIG Center (online) / toitutor.in.th — 2022 to present

- **What it is:** two online platforms that coach school students for Thailand's national programming olympiads. **Founded and run** since 2022; **coached hundreds of students** in live classes and one to one.
- **Courses I wrote and teach: COM300**, C++ and algorithms (9 chapters, about 270 pages, **128 recorded lessons**), and **COM100**, maths and **Python** for the olympiad exam (10 chapters with answer keys).

Algorithms and data structures · Competitive programming · Curriculum design · Teaching · C++ · Python

R&D Intern, AI research team

Amity Solutions / Bangkok, Thailand — Jul to Aug 2023

- **What it is:** the AI research and development team, building chatbots on **large language models (LLMs)**.
- **From paper to pipeline:** read recent **LLM research papers**, applied their ideas in the team's pipeline and **tested them end to end**, from the user's question to the answer.
- **RAG chatbots:** take the question, **search the knowledge base**, give the best-matching documents to the LLM with the question, and return its answer. Also data analysis and **prompt engineering**.

Large language models · Retrieval-augmented generation (RAG) · Applying research papers · End-to-end testing · Prompt engineering · Data analysis

04 Awards and certifications

2022 ● **National Champion: 1st place in Thailand and Gold Medal**
Thailand Olympiad in Informatics (TOI18)

2025 **1st place overall**
Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

2022 **Bronze Medal and Popular Vote**
SCIUS Forum 2022, Automated Cat Feeder

2026 **Best Paper Award**
IEEE JCSSE 2026, Bangkok, for U-HAP (second author of 4)

2023 to 2027 **Full scholarship**
4-Year Continuing Academic Outstanding Student Program, SIIT

CERTIFICATIONS ✓ **ISC2 Certified in Cybersecurity (CC)** 2025 · planned **CompTIA Security+** Exam planned Dec 2026

05 Projects

TU Alert, Novel Nest, Arduino retro console, Automated Cat Feeder, Velour Mall.

06 Research

Research supervisor: Prof. Somchart Fugkeaw, School of ICT, SIIT.

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8. One question links most of them: can a security check be cheap enough to run on every request, and still be provable afterwards?

FEDAC

● Under review · IEEE TDSC

ZERO-TRUST ACCESS CONTROL

First author of 4

Authorization-as-Result: decide once whether a caller may use a service, then hand over a **short-lived token for that one service**, so services do not re-run policy checks on every call.

- **Risk-adaptive sign-in:** the more sensitive the service, the stronger the proof, in four tiers from single sign-on to a **zero-knowledge proof**; a **risk score** from device, location and behaviour must stay under a threshold.
- **Many policy types, one template:** different kinds of access-control policy compile into one runtime template: what is allowed, under which conditions, which sign-in tier, for how long.
- **Privacy across services:** a user proves only the attributes a service needs, so services cannot link the user across each other.
- **Smart renewal:** a second score (policy freshness, trust, time left) chooses to extend a token, rebuild it, or ask for a new sign-in.
- **Verifiable governance:** policies on IPFS with Merkle roots on a **blockchain**, kept off the request path; a change revokes only the affected tokens.

Zero trust · Zero-knowledge proofs · Service mesh · Blockchain audit · Cloud benchmarking · Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

RESULT **1.4 ms** median authorization latency on AWS EKS, about 7,100 requests per second (paper under review)

U-HAP

● Presented · IEEE JCSSE 2026 · Best Paper Award

ZERO-TRUST ACCESS CONTROL

Second author of 4 · presented the paper

Authorization as a compiler problem: every Kubernetes access rule becomes part of **one graph**, and a decision becomes one question: can this user reach this resource?

- **One model for every rule type:** RBAC, ABAC, ACL, role hierarchy and **deny rules** share one meaning, without rewriting policies into a standard.
- **Compiled ahead of time** into indexes per resource and action, so nothing scans the whole policy set when a request arrives.
- **Two-level pruning:** skip rule types that are not in use, then check only the rules that match the caller's roles and attributes.
- **Deterministic:** conflicts follow explicit rules (deny wins), and an automated check confirms the same decisions as OPA.

Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Graph models · Python · Kubernetes webhook · OPA (baseline)

RESULT **~2.7× faster** policy updates reach a first decision vs OPA at 2,000 policies, with identical decisions

ZKAudit

● Accepted · KST 2027

ZERO-KNOWLEDGE AUDIT

First author

Bind who may audit to exactly what is audited: an auditor proves in **zero knowledge** that it holds a valid, properly scoped credential, and its admission is tied to the exact log state it then checks.

- **Private admission:** the credential and its attributes stay hidden; a valid proof earns a signed token for one domain and one time period.
- **No central bottleneck:** proof checks are spread over independent verifier shards and run in batches, so many auditors can be admitted at once.

Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Privacy · Rust · Groth16 zk-SNARKs · AWS EC2

Threshold-gated encrypted SIEM

● In preparation

SECURITY OPERATIONS

First author

Catch an attack that crosses companies without anyone sharing readable logs: the platform correlates logs encrypted under each company's own key.

- **k-of-n gate:** a trail may cross into another company's logs only when several companies independently reported the same signal, committed before anyone asked.

SIEM and threat detection · Applied cryptography · Threshold schemes · Privacy across companies · Python · Detection pipeline · LANL enterprise dataset

Post-quantum authorization across Kubernetes clusters

● Submitted

POST-QUANTUM SECURITY

First author

Verify the decision, do not trust the controller: each cluster checks for itself that an access decision used **every** allow and deny rule that applies, under **post-quantum** signatures.

- **Scope-complete:** the allow and deny rule sets are committed when compiled, so a controller cannot quietly leave out a deny rule.
- **Target-approved mapping:** a role from one cluster never silently becomes a stronger role in another; the target cluster approves the mapping.
- **Pay the post-quantum cost once:** a verified decision becomes a chain of light session capabilities, with batched auditing.
- **Precise updates:** when a policy, mapping or revocation changes, only the affected cached decisions are dropped; ledger commitments per epoch stop rollbacks.

Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Distributed ledgers · Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

06 Research — continued

Self-healing fog computing for industrial IoT

• Submitted

RESILIENT CLOUD AND EDGE — First author

Fog clusters that cooperate like a swarm, with no central controller: scheduling, load sharing and recovery are solved together, not as separate problems.

- **Shared experience:** clusters swap small signals about load, resources, links and readiness that fade and are reinforced over time, like ant trails, and schedule by trust plus that experience.
- **Predictive self-healing:** state is copied ahead of time and recovery plans are ready, so a failed coordinator is replaced quickly without a global re-plan.

Fog and edge computing · Industrial IoT · Fault tolerance · Homomorphic encryption · Python · gRPC · Paillier · Merkle proofs · AWS EC2

MoCA: AI containment

• In preparation

AI-AGENT SECURITY — First author

Assume the prompt injection succeeds, and make sure it cannot become authority: an AI agent's rights are checked when it acts, not when it plans.

- **Tools are labelled too:** a tool definition is an offer of authority; an unverified tool can still do low-risk work but can never raise what an action may do.
- **Delegation ceiling:** an agent never holds more than the user who delegated to it, and authority never grows along an agent-to-agent chain.
- **Two checkpoints:** the approved tool set is bound into the user's signature, and checks run at a **gateway and at the resource**, so code-running agents cannot go around them.

AI-agent security · Delegation and revocation · Prompt injection · Python · MCP tool gateway · Attack test harness

Fresh answers from elastic multi-cloud services

• In preparation

RESILIENT CLOUD AND EDGE — First author

Scheduling that knows whether data is fresh: a stateful service scales across clouds without ever quietly serving old data.

- **Freshness is proven, not guessed:** small Merkle proofs show that a replica is up to date, with no blockchain on the request path.
- **Verification first:** keep only replicas proven fresh, then choose by data locality, network distance, queue length and GPUs; a new replica joins only after it has synced.
- **Recovery as a cost trade-off:** reroute to a fresh peer instead of a cold restart when that is cheaper.

Multi-cloud systems · Elastic scaling · Freshness proofs · Python · Sparse Merkle trees · Multi-cloud deployment

ZK-ALBA

• In preparation · journal paper

ZERO-KNOWLEDGE AUDIT — First author

Extends ZKAudit: proves that the auditor may audit **right now** and that the evidence it gets is complete and intact, in one design.

- **Proof bound to its context:** each proof is tied to the authorization state, the exact log commitment, the audit target and the requester's key, so it cannot be reused.
- **Load-aware verifier shards** share one single-use registry, so the same admission cannot be replayed on another shard.
- **Logs stay encrypted:** the auditor receives only the ciphertexts for its target and checks origin, inclusion and completeness on them, never seeing plaintext or keys.

Zero-knowledge proofs · Scoped audit · Load balancing · Replay protection · Rust · Sharded verifiers · AWS EC2

07 Skills

Programming	**Python** · **C++** (main) · Go · Rust · TypeScript · C · Java · Bash · SQL
Used in my research	Zero Trust · Kubernetes (RBAC, ABAC, webhooks) · AWS (EKS, EC2) · Docker · Envoy and Istio · OPA / Rego · IPFS · Hyperledger Fabric
AI agents and AI security	LLM agent pipelines · MCP gateway · Prompt-injection testing · Prompt engineering · NLP chatbots
Shipping and operations	GitHub Actions CI · Semgrep · gitleaks · Playwright · pm2 on Linux · Backups and restore
Cryptography	zk-SNARKs (Groth16) · Merkle commitments · ML-DSA · ML-KEM
Security tools (labs and CTFs)	Nmap · Wireshark · Burp Suite · Metasploit · OSINT · OWASP Top 10
Also familiar with	Splunk and Wazuh (SOC labs) · Azure and AWS basics (DataCamp)
Languages	Thai (native) · English (fluent) · Chinese (basic)

08 References

People I have worked with, who agreed to be named, with their work contacts.

Apichai Wongchindakhun
Chief Information Security Officer (CISO)
Allianz Ayudhya

Mentored me during my information security internship (Jun to Jul 2026), where I built and presented two AI-agent prototypes for third-party risk.

Internship mentor · LinkedIn

Prof. Somchart Fugkeaw
Professor, School of ICT
SIIT, Thammasat University

Supervises my research and co-authors my papers, including FEDAC (under review, IEEE TDSC) and ZKAudit (accepted, KST 2027).

Research supervisor · somchart@siit.tu.ac.th

Pachara
Co-developer, TU Alert
SIIT, Thammasat University

Builds TU Alert with me for Thammasat University.

Project partner · Contact through me

Let's talk

Looking for an internship from January 2027 and full-time roles from May 2027: security engineering, cloud security or AI security.

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